

Scale of the universe

The difference in size between the smallest and biggest things in the universe is unimaginably vast—from subatomic particles to galaxies.

No one knows how big the universe is, but it has been expanding since it formed in the Big Bang 13.7 billion years ago. The distances are so great that cosmologists measure them in terms of light-years—the distance light moves in space in a year, which is equal to 6 trillion miles (9.5 trillion kilometers)—and parts of the universe are billions of light-years apart.

Quark

Too small to measure, different kinds of quarks are the subatomic particles that make up protons and neutrons.

Proton

A particle in an atomic nucleus that carries a positive charge.

Neutron

A particle in an atomic nucleus that carries no charge.

Carbon atom

With six electrons orbiting a nucleus of six protons and six neutrons, a carbon atom is less than a billionth of a meter across.

Limestone rock

Solid mixtures of billions of tiny fossilized shells and mineral fragments make up limestone rock, containing calcium carbonate—a compound that has atoms of calcium, carbon, and oxygen.

Nearest neighbor

Andromeda—the closest major galaxy to our Milky Way—is 2.5 million light-years from Earth.

ANDROMEDA

TRIANGULUM

LOCAL GROUP

MILKY WAY

Group cluster

Our Local Group is within a supercluster that is about 110 million light-years in diameter.

Supercluster

Clusters of galaxies span a region of space ten times bigger than the Local Group. Such a supercluster can contain tens of thousands of galaxies. Our Milky Way is within the Virgo Supercluster. Scientists think there are about 10 million superclusters in the observable universe.

Local Group

The Milky Way is part of a so-called Local Group of about 50 galaxies that stretch across 10 million light-years of space—that's 100 times the diameter of our Milky Way. Galaxies are millions of times farther apart than the stars that are in each one. Andromeda is the biggest galaxy in our Local Group—most others are much smaller.